

Day 21 Problem Set

Geometry Summer Credit | Surface Area, Volume, and Cross-Sections

Name:	Date:	Period:
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Directions

Identify the solid first. Label surface area in square units and volume in cubic units.

Part A: Prisms and Cylinders

1.	Find the volume of a rectangular prism with length 14, width 6, and height 5.
2.	Find the surface area of a rectangular prism with length 9, width 4, and height 7.
3.	A triangular prism has a triangular base with area 30 and prism length 12. Find the volume.
4.	A cylinder has radius 7 and height 10. Find the volume in terms of pi.
5.	A cylinder has radius 3 and height 8. Find the surface area in terms of pi.
6.	A cylinder has diameter 12 and height 15. Find the volume in terms of pi.
7.	Error analysis: A student gives a volume answer in square units. Explain the mistake.

Part B: Pyramids, Cones, and Spheres

8. A square pyramid has base side 10 and height 9. Find the volume.

9. A cone has radius 4 and height 12. Find the volume in terms of π .

10. A sphere has radius 6. Find the volume in terms of π .

11. A sphere has radius 5. Find the surface area in terms of π .

12. A cone has radius 5 and slant height 13. Find its surface area in terms of π .

13. A square pyramid has base side 8 and slant height 6. Find its surface area.

14. A cone has volume 96π and radius 6. Find its height.

Day 21 Problem Set

Continued | Cross-Sections and Spiral Review

15. Name the cross-section made by slicing a cylinder parallel to its base.

16. Name one possible cross-section made by slicing a rectangular prism.

17. A right rectangular prism is sliced parallel to one face. What kind of cross-section can result?

18. Spiral review: A trapezoid has bases 18 and 30 and height 7. Find its area.

19. Spiral review: A circle has radius 9 and central angle 80 degrees. Find arc length.

20. Application: A shipping box is 18 in by 12 in by 10 in. Find the amount of cardboard needed to cover the outside, ignoring overlaps.

21. Challenge: A cylindrical can has radius 4 cm and height 12 cm. Which is greater, its volume or surface area? Explain why the units make direct comparison tricky.